

## / Doc changelog

### Create changelog fragment

```
- uses: ansys/actions/doc-changelog@v10.3
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

*Ref:* [Create changelog fragment](#)

## / Doc deploy dev

### Deploy developers documentation

```
- name: Deploy the latest documentation
  uses: ansys/actions/doc-deploy-dev@v10.3
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    cname: <library>.docs.pyansys.com
    token: ${{ secrets.GITHUB_TOKEN }}
```

*Ref:* [Deploy developers documentation](#)

## / Doc build

### Building library documentation

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@v10.3
```

*Ref:* [Building library documentation](#)

### Building library documentation that using xvfb

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@v10.3
  with:
    requires-xvfb: true
```

*Ref:* [Building library documentation that using XVFB](#)

### Installing additional system dependencies for building documentation

```
- name: Run Ansys documentation building action
  uses: ansys/actions/doc-build@v10.3
  with:
    dependencies: graphviz mermaid-cli
```

*Ref:* [Installing additional system dependencies for building documentation](#)

## / Check licenses

### Check library dependencies ship with valid licenses

```
- name: Validate third party licenses
  uses: ansys/actions/check-licenses@v10.3
  with:
    python-version: ${{ env.MAIN_PYTHON_VERSION }}
```

*Ref:* [Check library dependencies ship with valid licenses](#)

## / Hk automerge prs

### Automerging dependabot and pre-commit.ci prs

```
- name: Automerging dependabot and pre-commit.ci PRs
  uses: ansys/actions/hk-automerge-prs@v10.3
  with:
    approver: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    approver-token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

*Ref:* [Automerging dependabot and pre-commit.ci PRs](#)

## / Hk package clean except

### Perform versions cleanup - except certain tags

```
- name: Perform versions cleanup - except certain tags
  uses: ansys/actions/hk-package-clean-except@v10.3
  with:
    package-name: mypackage
    tags-kept: latest, latest-unstable
    token: ${{ secrets.GITHUB_TOKEN }}
```

*Ref:* [Perform versions cleanup - except certain tags](#)

## / Doc deploy changelog

### Update changelog for new tag

```
- uses: ansys/actions/doc-deploy-changelog@v10.3
  with:
    bot-email: ${{ secrets.PYANSYS_CI_BOT_EMAIL }}
    bot-user: ${{ secrets.PYANSYS_CI_BOT_USERNAME }}
    token: ${{ secrets.PYANSYS_CI_BOT_TOKEN }}
```

*Ref:* [Update CHANGELOG for new tag](#)

## / Build ci wheels

### Build a c-extension library

```
- name: Build a C-extension library wheel artifacts
  uses: ansys/actions/build-ci-library@v10.3
```

*Ref:* [Build a C-extension library](#)

## / Tests pytest

### Optimized testing with different operating systems and python versions

```
- name: Login into the GitHub container registry
  uses: docker/login-action@f4ef78c080cd8ba55a85445d5b3
  with:
    password: ${{ secrets.GITHUB_TOKEN }}
    registry: ghcr.io
    username: ${{ secrets.GH_USERNAME }}
- env:
  LICENSE_SERVER: ${{ secrets.LICENSE_SERVER }}
  name: Pull, launch and verify desired Ansys service
  run: '...'
- name: Run pytest with desired markers and extra argument
  uses: ansys/actions/tests-pytest@v10.3
  with:
    pytest-extra-args: ${{ matrix.cfg.extra-args }}
    pytest-markers: ${{ matrix.cfg.markers }}
    requires-xvfb: true
```

*Ref:* [Optimized testing with different operating systems and Python versions](#)

### Testing library with different operating systems and python versions

```
- name: Run pytest
  uses: ansys/actions/tests-pytest@v10.3
  with:
    pytest-extra-args: --cov=ansys.<library> --cov-repo
    pytest-markers: -k 'mocked'
```

*Ref:* [Testing library with different operating systems and Python versions](#)

## / Check pr title

### Check pull-request title follows conventional commits

```
- uses: ansys/actions/check-pr-title@v10.3
  with:
    token: ${{ secrets.GITHUB_TOKEN }}
```

*Ref:* [Check pull-request title follows conventional commits](#)

## / Docker style

### Docker style

```
- uses: ansys/actions/docker-style@v10.3
  with:
    directory: docker
    error-level: 1
    recursive: true
```

*Ref:* [Docker style](#)

## / Code style

### Running code style checks

```
- name: Run PyAnsys code style checks
  uses: ansys/actions/code-style@v10.3
```

*Ref:* [Running code style checks](#)

## / Hk tag repository version

### Tag repository version

```
- uses: actions/checkout@v6
  with:
    fetch-depth: 0
    fetch-tags: true
- env:
  SEMANTIC_VERSION: ${ github.event.release.tag_name }
  id: major_minor_version
  name: vX.Y version from tag
  run: 'MAJOR_VERSION=$(echo ${SEMANTIC_VERSION} | cut -d 'v' -f 2 | cut -d '.' -f 1)

  MINOR_VERSION=$(echo ${SEMANTIC_VERSION} | cut -d 'v' -f 2 | cut -d '.' -f 2)

  echo "MAJOR_MINOR_VERSION=v${MAJOR_VERSION}.${MINOR_VERSION}" >> $GITHUB_OUTPUT
```

```

  shell: bash
- name: Update latest vX and vX.Y tag
  uses: ansys/actions/hk-tag-repository-version@v10.3
  with:
    gh-token: ${ secrets.GITHUB_TOKEN }
    latest-branch-name: release/${ steps.major_minor
    }
```

*Ref:* [Tag repository version](#)

#### Tag repository version

```
- uses: actions/checkout@v6
  with:
    fetch-depth: 0
    fetch-tags: true
- name: Update latest vX and vX.Y tag
  uses: ansys/actions/hk-tag-repository-version@v10.3
  with:
    gh-token: ${ secrets.GITHUB_TOKEN }
    release-version: ${ github.event.release.tag_name
    }
```

*Ref:* [Tag repository version](#)

## / Release pypi public

#### Release to public pypi

```
- name: Release to the public PyPI repository
  uses: ansys/actions/release-pypi-public@v10.3
  with:
    library-name: ansys-<product>-<library>
    twine-token: ${ secrets.PYPI_TOKEN }
    twine-username: __token__
```

*Ref:* [Release to public PyPI](#)

## / Release pypi private

#### Release to private pypi

```
- name: Release to the private PyPI repository
  uses: ansys/actions/release-pypi-private@v10.3
  with:
    library-name: ansys-<product>-<library>
    twine-token: ${ secrets.PYANSYS_PYPI_PRIVATE_PAT
    twine-username: __token__
```

*Ref:* [Release to private PyPI](#)

## / Check vulnerabilities

#### Check library vulnerabilities

```
- uses: ansys/actions/check-vulnerabilities@v10.3
  with:
    dev-mode: ${ github.ref != 'refs/heads/main' }
    python-package-name: ansys-<product>-<library>
    python-version: ${ env.MAIN_PYTHON_VERSION }
    token: ${ secrets.PYANSYS_CI_BOT_TOKEN }
```

*Ref:* [Check library vulnerabilities](#)

## / Hk migrate fork pr

#### Migrate fork pr to main repository

```
- name: Migrate fork PR
  uses: ansys/actions/hk-migrate-fork-pr@v10.3
  with:
    bot-email: ${ secrets.PYANSYS_CI_BOT_EMAIL }
    bot-username: ${ secrets.PYANSYS_CI_BOT_USERNAME }
    comment-id: ${ github.event.comment.id }
    github-token: ${ secrets.PYANSYS_CI_BOT_TOKEN }
    pr-number: ${ github.event.issue.number }
    team-slug: pyansys-maintainers
    user-triggering: ${ github.event.comment.user.login
    }
```

*Ref:* [Migrate fork PR to main repository](#)

## / Doc deploy stable

#### Deploy stable documentation

```
- name: Deploy the stable documentation
  uses: ansys/actions/doc-deploy-stable@v10.3
  with:
    bot-email: ${ secrets.PYANSYS_CI_BOT_EMAIL }
    bot-user: ${ secrets.PYANSYS_CI_BOT_USERNAME }
    cname: <library>.docs.pyansys.com
    token: ${ secrets.GITHUB_TOKEN }
```

*Ref:* [Deploy stable documentation](#)

## / Build wheelhouse

#### Build wheelhouse for latest python versions

```
- name: Build a wheelhouse of the Python library
  uses: ansys/actions/build-wheelhouse@v10.3
  with:
    library-name: <ansys-product-library>
    operating-system: ${ matrix.os }
    python-version: ${ matrix.python-version }
```

*Ref:* [Build wheelhouse for latest Python versions](#)

## / Hk package clean untagged

#### Perform untagged versions cleanup

```
- name: Perform untagged versions cleanup
  uses: ansys/actions/hk-package-clean-untagged@v10.3
  with:
    package-name: mypackage
    token: ${ secrets.GITHUB_TOKEN }
```

*Ref:* [Perform untagged versions cleanup](#)

## / Build library

#### Build library basic example

```
- name: Build library source and wheel artifacts
  uses: ansys/actions/build-library@v10.3
  with:
    library-name: ansys-<product>-<library>
```

*Ref:* [Build library basic example](#)

## / Check actions security

#### Check actions security

```
- uses: ansys/actions/check-actions-security@v10.3
  with:
    auditing-level: high
    generate-summary: true
    token: ${ secrets.GITHUB_TOKEN }
    trust-ansys-actions: true
```

*Ref:* [Check actions security](#)

## / Doc deploy pr

#### Deploy pr documentation

```
- uses: ansys/actions/doc-deploy-pr@v10.3
  with:
    bot-email: ${ secrets.PYANSYS_CI_BOT_EMAIL }
    bot-user: ${ secrets.PYANSYS_CI_BOT_USERNAME }
    cname: ${ env.DOCUMENTATION_CNAME }
    maximum-pr-doc-deployments: 3
    token: ${ secrets.GITHUB_TOKEN }
```

*Ref:* [Deploy PR documentation](#)

## / Release pypi test

#### Release to test pypi

```
- name: Release to the test PyPI repository
  uses: ansys/actions/release-pypi-test@v10.3
  with:
    library-name: ansys-<product>-<library>
    twine-token: ${ secrets.PYANSYS_PYPI_TEST_PAT
    twine-username: __token__
```

*Ref:* [Release to test PyPI](#)

## / Release github

#### Release to github

```
- name: Release to GitHub
  uses: ansys/actions/release-github@v10.3
  with:
    library-name: ansys-<product>-<library>
```

*Ref:* [Release to GitHub](#)

## / Doc style

#### Running documentation style checks

```
- name: Running documentation style checks
  uses: ansys/actions/doc-style@v10.3
  with:
    token: ${ secrets.GITHUB_TOKEN }
```

*Ref:* [Running documentation style checks](#)